

AetherNet Guardian Down
**Investigating Server Performance Anomalies Using Principal
Component Analysis (PCA)**

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09 June 2026

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Executive Summary

AetherNet Cloud Solutions observed unusual behaviour across several servers within its European infrastructure. Although no major outage occurred, increasing disk activity, higher temperatures, and network performance issues raised concerns regarding system reliability. To investigate the root cause of these anomalies, Principal Component Analysis (PCA) was applied to a telemetry dataset collected from three server clusters over a ten-day period.

The analysis identified three main patterns within the data. The first principal component (PC1) was strongly associated with storage and hardware-related variables such as disk errors, disk queue depth, disk activity, and temperature. The second principal component (PC2) represented network-related behaviour through latency, packet loss, CPU usage, and memory usage. The third component (PC3) reflected overall operational stability through variables such as memory page faults and power consumption.

The results indicate that performance degradation likely began with storage-related issues before affecting other areas of the infrastructure. Cluster 3 showed the strongest signs of deterioration and should be considered the highest priority for further investigation.

1. Introduction

AetherNet Cloud Solutions provides cloud services to organisations that depend on reliable and secure computing infrastructure. Recently, monitoring systems detected unusual activity across several servers. While the systems remained operational, the growing number of performance anomalies suggested that underlying issues were developing.

The purpose of this project was to identify the main factors contributing to the observed behaviour and determine whether specific clusters were experiencing abnormal conditions. To achieve this, Principal Component Analysis (PCA) was used to reduce the complexity of the telemetry dataset and identify the most important sources of variation.

The objectives of this analysis were:

- Prepare and clean the telemetry dataset.
- Explore relationships between telemetry variables.
- Identify the principal factors affecting system performance.
- Determine which cluster is most affected.
- Provide recommendations for further investigation.

2. Data Preparation and Exploratory Data Analysis

The telemetry dataset was inspected for missing values and unnecessary variables before analysis. Variables that did not contribute meaningful information to PCA were removed, while missing numerical values were replaced using median imputation. All numerical variables were then standardised to ensure equal contribution during the PCA process.

The correlation matrix shown in **Figure 1** indicates that several storage-related variables are positively correlated, particularly disk throughput, disk queue depth, and corrected disk errors. This suggests that these metrics describe similar underlying behaviour and supports the use of PCA as a dimensionality reduction technique.

Additional exploratory analysis was performed using cluster-level visualisations. **Figure 5** shows that Cluster 3 recorded higher levels of corrected disk errors compared with the other clusters. Similarly, **Figure 6** indicates slightly elevated temperatures within Cluster 3. These observations provide early evidence that storage and hardware-related issues may be developing within this cluster.

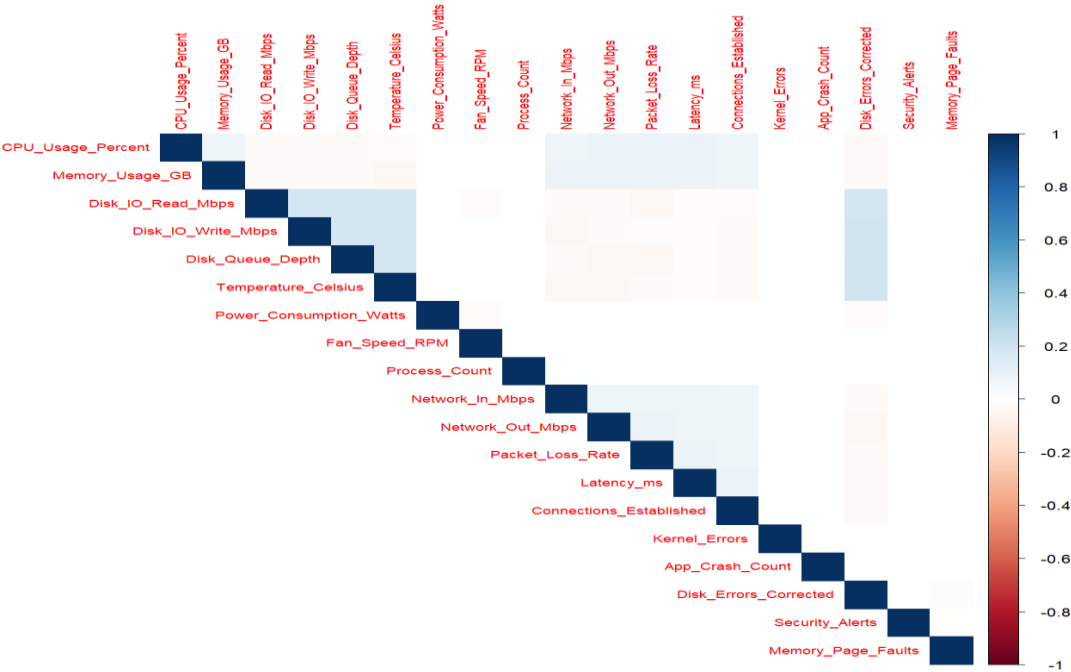


Figure 1: Correlation Matrix

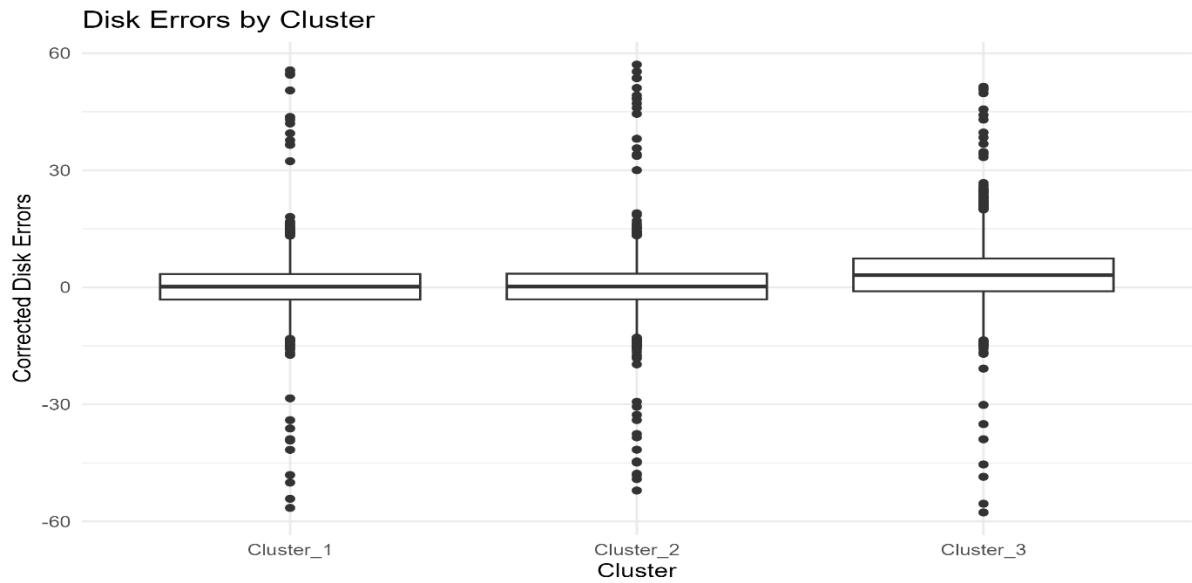


Figure 5: Disk Errors by Cluster

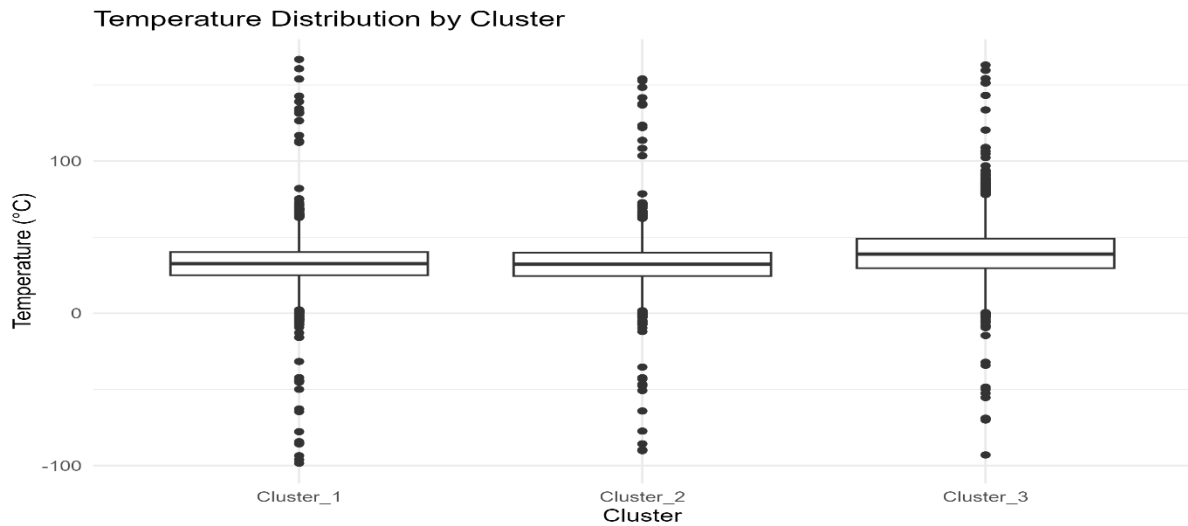


Figure 6: Temperature by Cluster

3. Principal Component Analysis Results

Principal Component Analysis was performed on the standardised telemetry variables. The amount of variance explained by each principal component is presented in **Table 1**.

Table 1. Variance Explained by Principal Components

Component	Variance Explained (%)
PC1	9.57
PC2	7.45
PC3	5.40
Total	22.41

The scree plot shown in **Figure 2** demonstrates that the first three components capture the most meaningful variation in the dataset. Although the total variance explained is relatively modest, this is expected because telemetry systems contain many independent operational metrics.

The variables contributing most strongly to PC1 are shown in **Table 2** and **Figure 3**.

Table 2. Top Variables Contributing to PC1

Variable	Loading
Disk_Errors_Corrected	0.418
Disk_Queue_Depth	0.411
Disk_IO_Write_Mbps	0.410
Temperature_Celsius	0.407

Variable Loading

Disk_IO_Read_Mbps 0.407

These variables are all associated with storage performance and hardware stress. Therefore, PC1 can be interpreted as a storage-related component representing increasing pressure on the storage subsystem.

PC2 was primarily influenced by latency, packet loss, CPU usage, and memory consumption, suggesting that it represents network performance. PC3 was associated with page faults, fan speed, and power consumption, indicating overall operational stability.

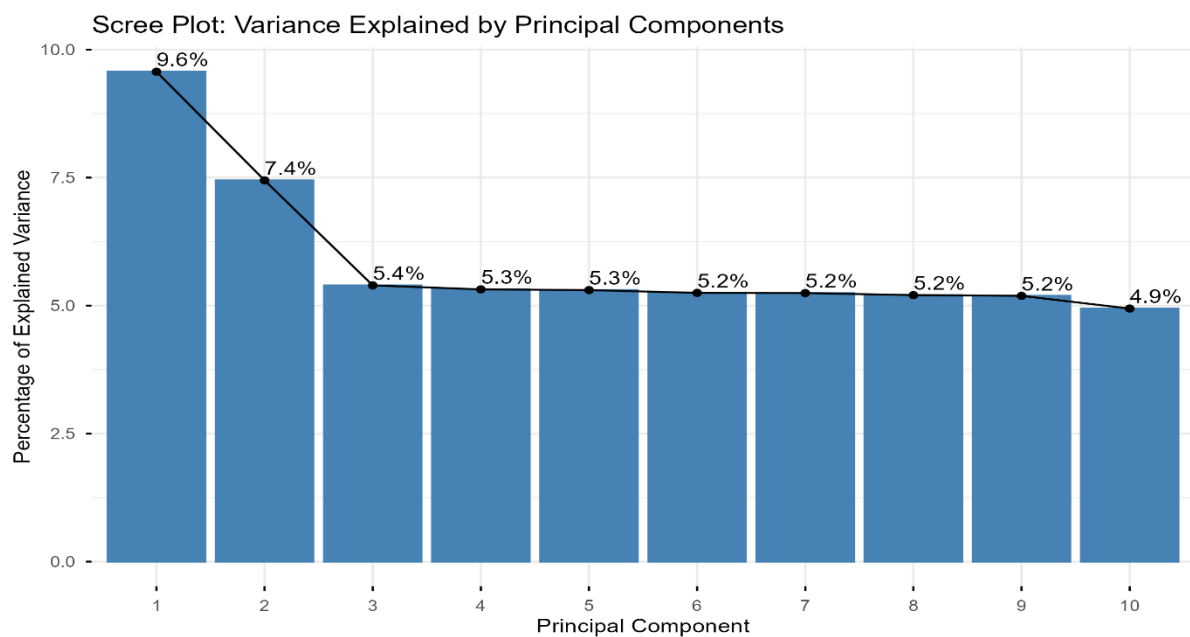


Figure 2: Scree Plot

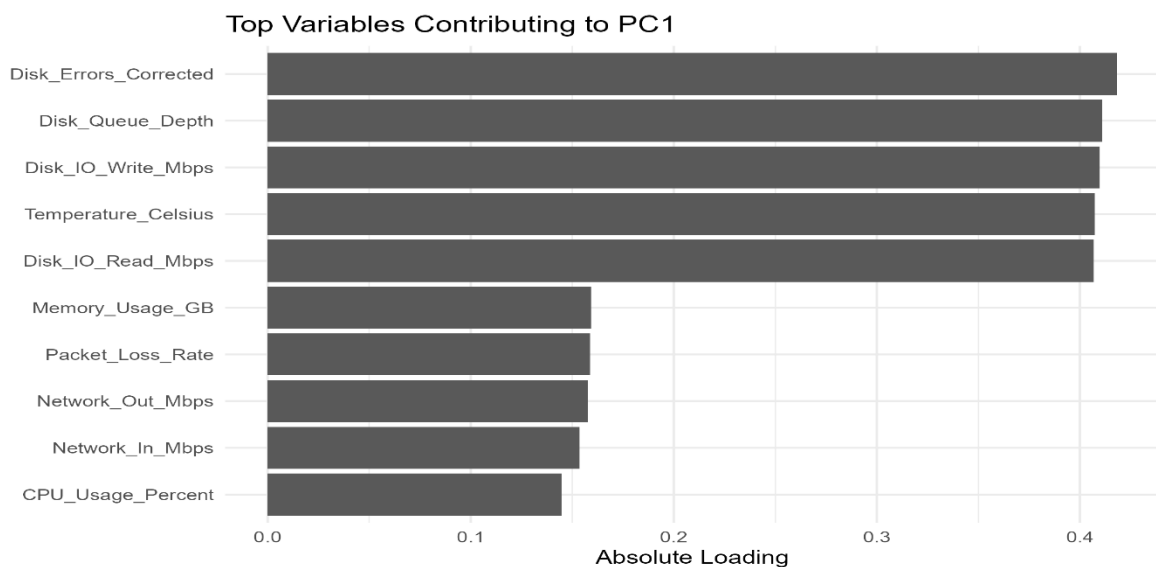


Figure 3: Top Variables Contributing to PC1

4. Temporal and Cluster Analysis

To understand how system behaviour changed over time, principal component scores were analysed across the three clusters.

Table 3. Cluster PCA Summary

Cluster Mean PC1

Cluster 1 0.441

Cluster 2 0.433

Cluster 3 -0.874

Although Cluster 1 recorded the highest average PC1 score, the temporal analysis reveals that Cluster 3 experienced the most significant deterioration over time.

As shown in **Figure 4**, PC1 scores for Cluster 3 decline noticeably after approximately 10 April. This pattern coincides with the increasing number of corrected disk errors observed during the same period. The results suggest that storage-related issues became progressively worse before influencing other aspects of system performance.

The combination of increasing disk errors, elevated temperatures, and deteriorating PCA scores indicates that Cluster 3 is the primary source of concern and should be prioritised for further investigation.

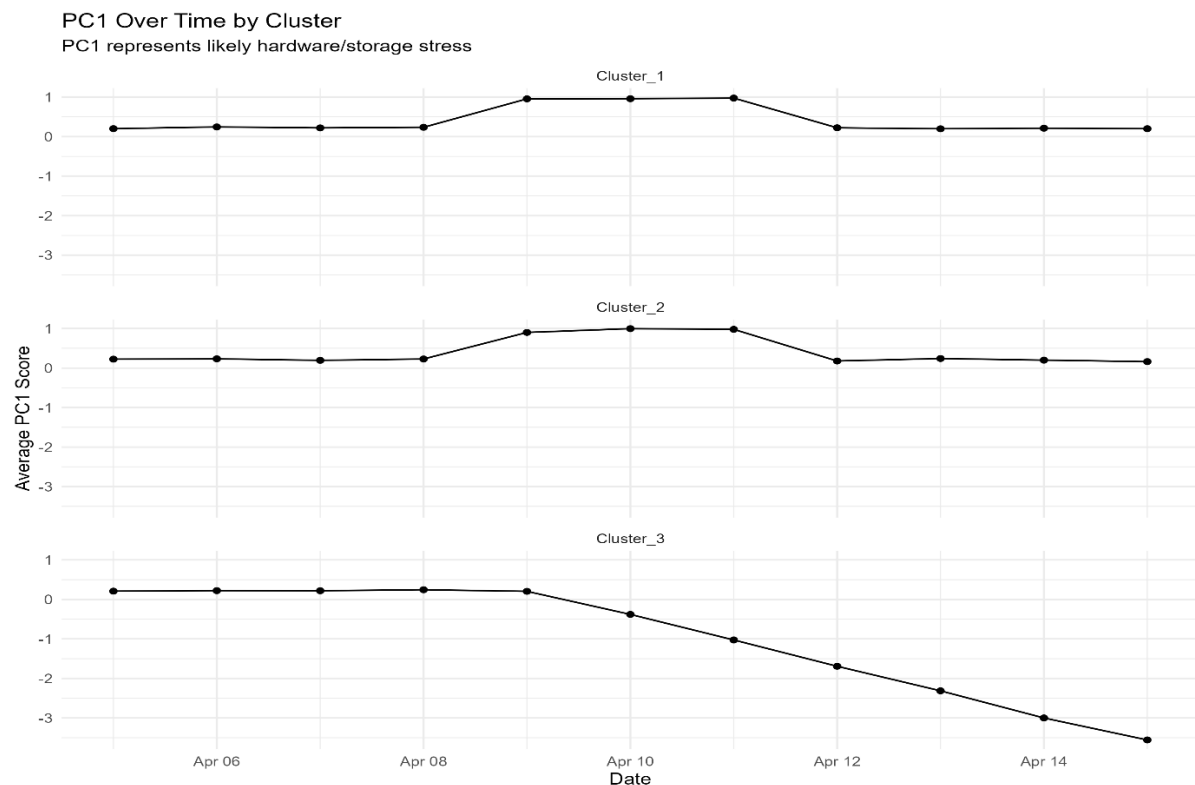


Figure 4: PC1 Over Time by Cluster

5. Recommendations

Based on the results of the analysis, the following actions are recommended:

High Priority

- Investigate storage infrastructure associated with Cluster 3.
- Review disk controller logs and corrected disk error records.
- Examine disk queue depth and I/O performance trends.
- Verify the effectiveness of cooling systems within affected servers.

Medium Priority

- Review latency and packet loss patterns to identify potential network bottlenecks.
- Investigate memory usage and page fault activity.
- Continue monitoring thermal conditions across all clusters.

Long-Term Improvements

- Implement PCA-based anomaly detection as part of the monitoring framework.
- Establish automated alert thresholds for storage-related metrics.
- Conduct periodic health checks on storage devices within high-risk clusters.

6. Conclusion

This study used Principal Component Analysis to investigate unusual behaviour within AetherNet's server infrastructure. The results suggest that storage-related issues are the primary driver of the observed anomalies. Disk errors, queue depth, disk activity, and temperature were identified as the most influential variables within the first principal component.

Temporal analysis further showed that Cluster 3 experienced the strongest deterioration, particularly after 10 April. The evidence suggests that storage subsystem stress developed first and subsequently affected broader system performance.

By addressing the storage-related issues identified in this report and implementing continuous monitoring strategies, AetherNet can reduce operational risk and improve the long-term reliability of its infrastructure.

7. Reference

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